
PHISHING EMAIL DETECTION & AWARENESS REPORT

Cybersecurity Task 2 - Future Interns

Organisation	Future Interns
Prepared by	Muhammad Ali
Scope	Phishing Email Detection & Awareness

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1. Executive Summary

This report provides a detailed examination of four phishing email samples analysed during a cyber security internship programme. The purpose of this report is to identify common phishing characteristics, assess email threat levels, and provide security awareness recommendations that can help individuals and enterprises in defending against phishing-related attacks.

Phishing continues to be one of the most prevalent cybersecurity threats impacting businesses and individuals worldwide. Cyber attackers frequently impersonate trusted companies via fraudulent emails to deceive users into revealing confidential information, opening harmful links, downloading malicious files, or making unauthorised financial transactions. The findings included in this report highlight the significance of user awareness, email verification practices, and preventive security measures for minimising the risk associated with phishing attacks.

2. Objectives

- Analyze phishing email samples
- Identify phishing indicators
- Classify email risks
- Understand phishing techniques
- Create security awareness guidance
- Recommend prevention measures

3. Scope & Methodology

3.1 Scope

- Analysis of phishing email samples collected for educational and awareness purposes
- Identification of common phishing indicators, including fake sender domains, suspicious links, urgency tactics, and credential requests
- Examination of social engineering methods used by attackers to deceive victims
- Assessment and classification of email threat levels according to identified phishing characteristics
- Evaluation of potential risks and consequences associated with phishing attacks on individuals and organisations
- Recommendation of preventive security measures and secure email handling practices

3.2 Tools Used

Tools	Purpose
MXToolbox	Email header analysis
Google Header Analyzer	Header inspection
Browser DevTools	Link inspection

3.3 Ethical Considerations

- Educational use only
- No hacking performed
- Passive analysis only
- No malicious activity

4. Common Phishing Indicators

No.	Phishing Indicator	Threat Level	Explanation
1.	Impersonal Greeting	Medium	Uses fake or modified domains to imitate trusted organisations
2.	Immediate Action Requests	High	Encourages immediate action through fear or deadlines
3.	Spoofed Sender Address	High	Uses non-personal greetings such as 'Dear User' or 'Customer'
4.	Suspicious URLs	High	Contains links directing users to fake or harmful websites
5.	Prevents Verification	High	Requests passwords, OTPs, or confidential account details
6.	Confidentiality Requests	High	Instructs users not to verify or discuss the message
7.	False Rewards & Offers	High	Promises prizes, financial rewards, or fake opportunities
8.	Credential Collection	Critical	Discourages users from contacting official support channels
9.	Fake Processing Fee Request	Critical	Requests upfront payments before providing services or rewards
10.	Executive Impersonation	Critical	Pretends to be a company executive requesting urgent transfers

5. Phishing Email Sample Analysis

5.1 Sample 1 - Amazon Account Verification Phishing Email

5.1.1 Email Type

Credential Phishing / Account Verification Scam

5.1.2 Suspicious Sender Address

security-alert@amazon-support-center.com

5.1.3 Email Subject

Important: Verify Your Amazon Account

5.1.4 Threat Severity Assessment

Confirmed Phishing Email – Critical Risk

5.1.5 Phishing Email Sample

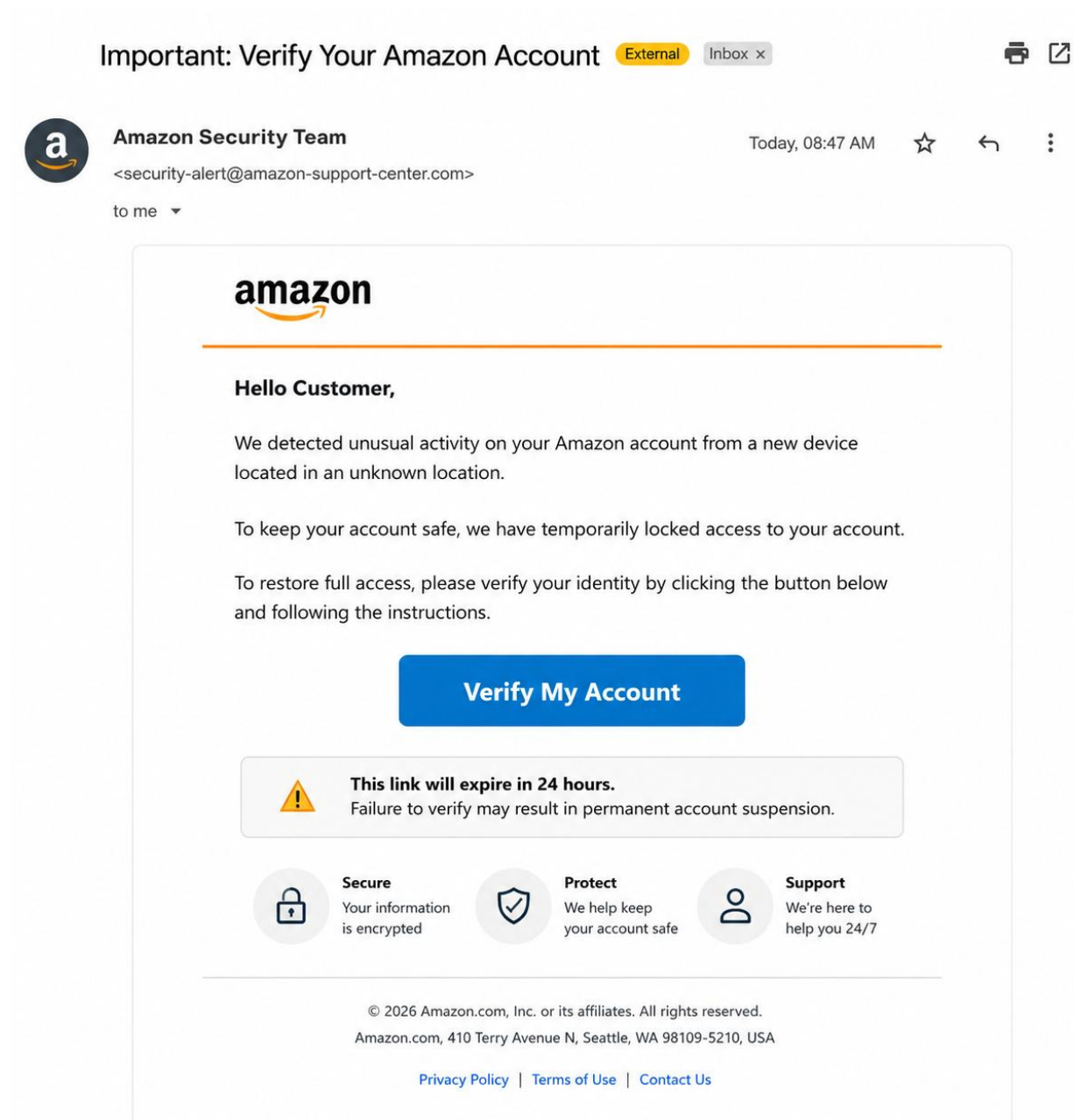


Figure 1: Amazon account verification phishing email sample

5.1.6 Email Header Analysis Using MXToolbox

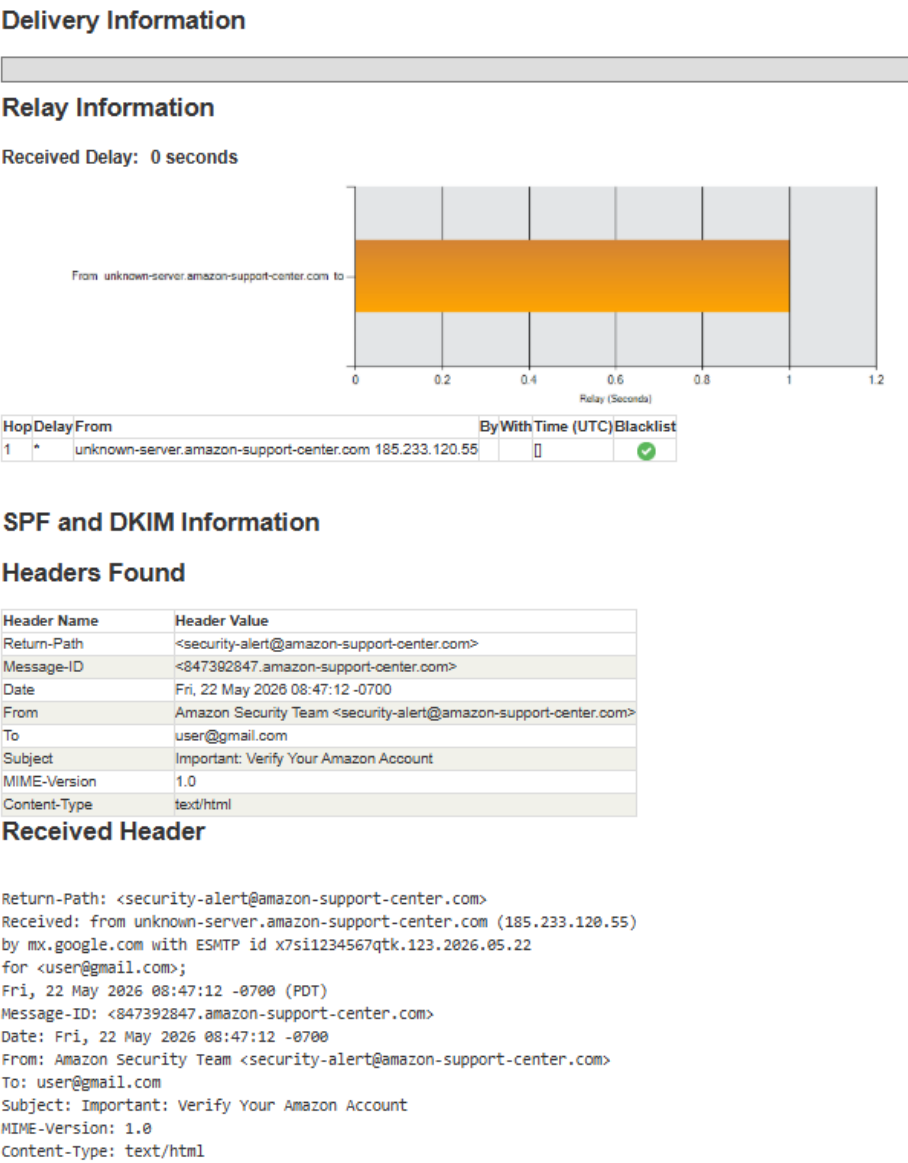


Figure 2: MXToolbox email header analysis results

Explanation

The MXToolbox header analysis detected suspicious sender infrastructure associated with the domain “amazon-support-center.com”, which does not match Amazon’s legitimate corporate

domains. The routing information showed that the email came from an unknown external server, implying possible spoofing activity commonly linked to phishing campaigns.

The analysis also revealed suspicious sender details, relay information, and external mail routing behaviour, which further confirmed the classification of the email as a phishing attempt. Additionally, the email attempted to create panic by requesting instant account verification, which is a common technique used by attackers to trick victims into revealing sensitive information.

5.1.7 Email Header Analysis Using Google Admin Toolbox

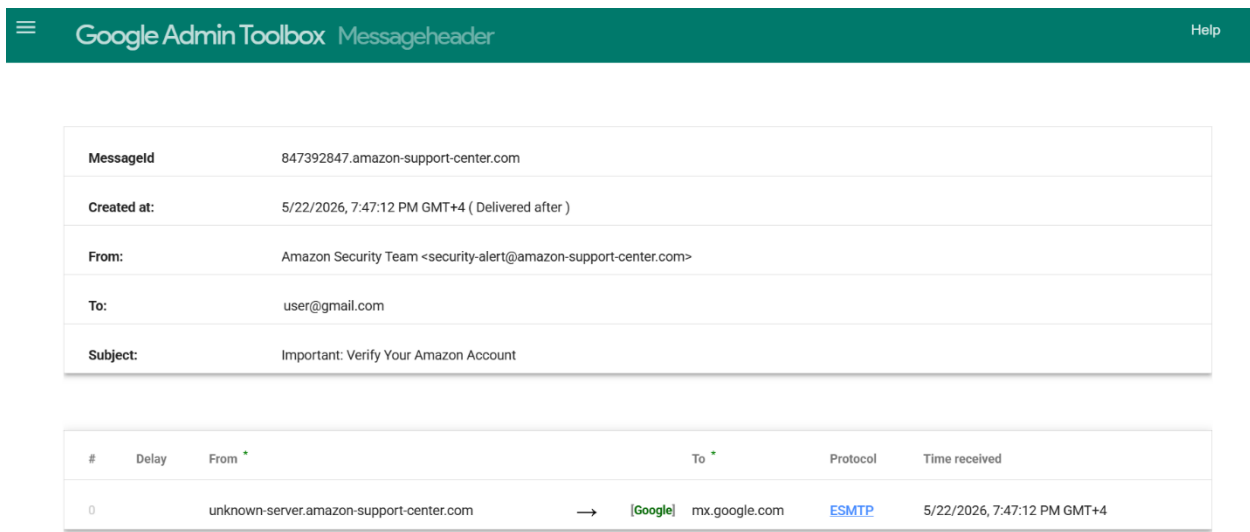


Figure 3: Google Admin Toolbox header analysis results

Explanation

The Google Admin Toolbox analysis found suspicious sender information associated with the domain “amazon-support-center.com”. The message routing details showed that the email originated from an unknown external server rather than Amazon’s official infrastructure.

The analysis also identified suspicious email delivery behaviour and spoofed sender details commonly linked with phishing campaigns. Moreover, the email employed urgency-based language asking for immediate account verification to manipulate the recipient into revealing sensitive information.

5.1.8 Identified Phishing Indicators

No.	Phishing Indicator	Threat Level	Explanation
1.	Impersonal Greeting	Medium	Uses “Hello Customer” instead of the recipient’s actual name
2.	Immediate Action Requests	High	Encourages urgent account verification within 24 hours
3.	Spoofed Sender Address	High	Uses unofficial domain “amazon-support-center.com” to imitate Amazon
4.	Suspicious URLs	High	Verification button may redirect users to a fraudulent login page
5.	Prevents Verification	High	Pressures users to act immediately without verifying legitimacy
6.	Credential Collection	Critical	Attempts to collect user login credentials and personal information
7.	Executive Impersonation	Critical	Uses Amazon branding and security-related messaging to appear legitimate

5.1.9 Phishing Attack Mechanism

The attacker impersonates Amazon by utilising a professionally crafted phishing email that mimics official Amazon branding and communication style. The email attempts to create panic by alerting the victim about suspicious account activity and possible account suspension.

The victim is urged to click the verification button, which would generally redirect users to a fraudulent login page created to steal account credentials and sensitive personal information.

5.1.10 Potential Security Impact

- Unauthorized access to customer accounts
- Exposure of personal and financial information
- Financial fraud and identity theft
- Loss of customer trust and account security

5.1.11 Security Recommendations

- Verify sender domains carefully before responding
- Avoid clicking suspicious links in emails
- Access Amazon services only through official websites
- Enable multi-factor authentication on accounts
- Report suspicious emails immediately to security teams

5.2 Sample 2 - Fraudulent Loan Approval Phishing Email

5.2.1 Email Type

Financial Fraud / Advance Fee Scam

5.2.2 Suspicious Sender Address

support@everestbank-secure.com

5.2.3 Email Subject

Great News! Your Personal Loan Has Been Approved

5.2.4 Threat Severity Assessment

Confirmed Phishing Email - Critical Risk

5.2.5 Phishing Email Sample

**Everest Banking**

Personal Banking Group
Reliable Banking. Lasting Relationships.



Great News!

Your Personal Loan Has Been Approved

We are happy to inform you that your loan application has been approved in principle.



LOAN APPROVED



What happens next?

To complete your loan disbursement, please verify your account and pay a one-time administrative fee **within 24 hours**.

Your Loan Summary

 Approved Amount \$25,000 USD	 Loan Type Personal Loan	 Interest Rate 3.45% Per Annum	 Tenure 36 Months
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Application ID	EB-PL-2026-0413-7786
Approval Date	April 13, 2026
Processing Fee	\$50 USD (One-time payment)



Action Required

Please verify your bank account details and pay the processing fee to activate your loan and release the funds.

Verify Account & Pay Fee

 This secure link will expire in **24 hours**.



Security Reminder

For your security, do not share this email or your reference number with anyone.

**Need Help?**
Call us 24/7
+1 (877) 555-2380

**Email Us**
loansupport@everestbank.com

**Visit Us**
www.everestbank.com

© 2026 Everest Banking. All rights reserved.
This is an automated message. Please do not reply to this email.

Figure 4: Fraudulent loan approval phishing email sample

5.2.6 Email Header Analysis Using MXToolbox

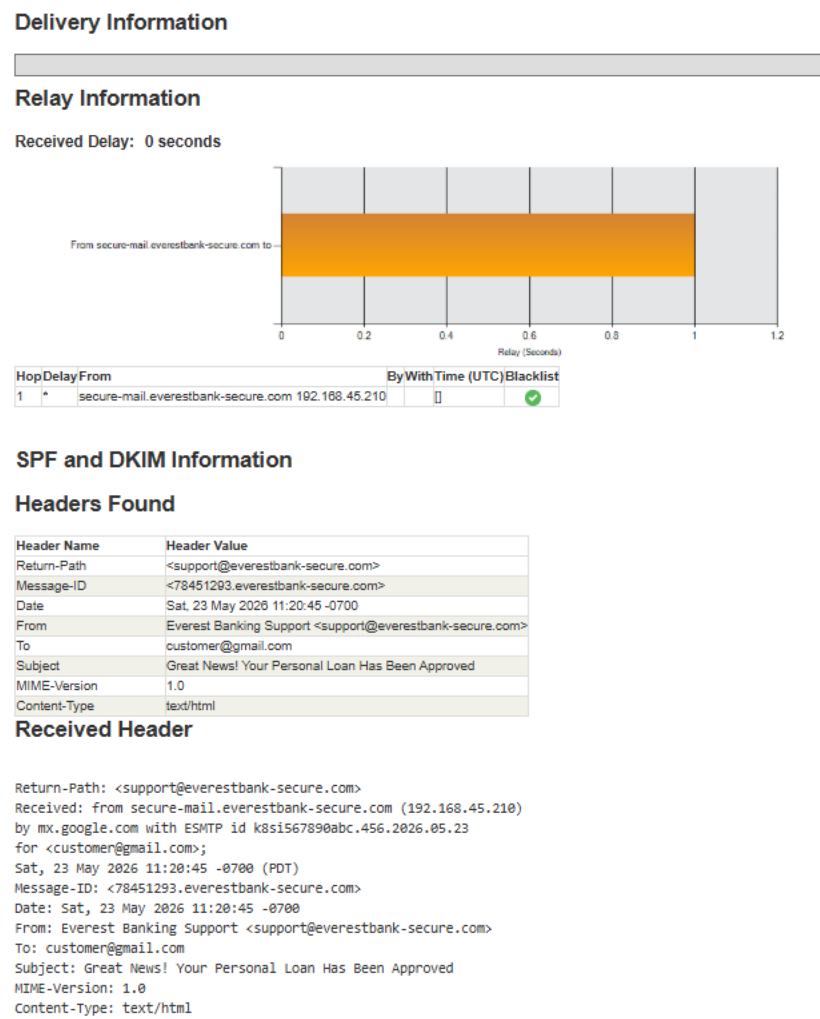


Figure 5: MXToolbox header analysis results for the fraudulent loan approval email

Explanation

The MXToolbox analysis detected suspicious sender infrastructure associated with the domain “everestbank-secure.com”, which does not align with the legitimate domain structure commonly used by official banking institutions. The routing information revealed that the email came from an unknown external server, indicating possible spoofing activity typically associated with phishing and financial fraud campaigns.

The analysis also uncovered suspicious sender details and external mail routing behaviour that further reinforce the classification of the email as a fraudulent phishing attempt. Furthermore, the email attempted to force the recipient into doing urgent verification and payment steps within a limited period.

5.2.7 Email Header Analysis Using Google Admin Toolbox

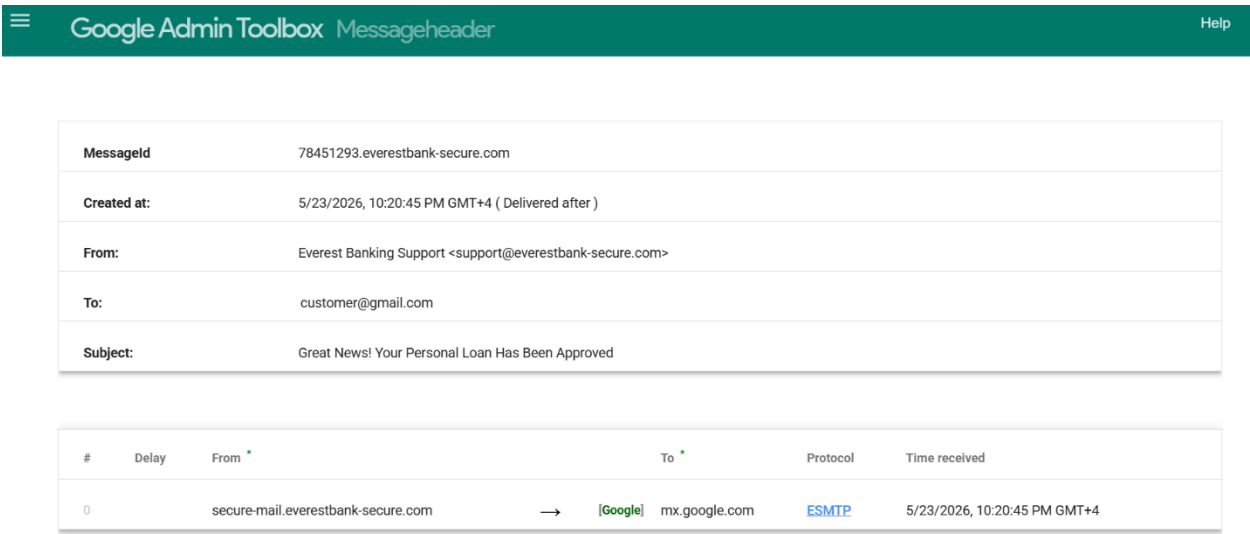


Figure 6: Google Admin Toolbox header analysis results for the phishing email

Explanation

The Google Admin Toolbox analysis uncovered suspicious sender domain information and external email routing details inconsistent with authentic banking communication systems. The sender address attempted to copy a trusted financial institution while using a deceptive external domain.

The analysis further revealed phishing characteristics commonly seen in financial fraud campaigns created to manipulate victims into disclosing sensitive banking information and unauthorised payments. The use of time-sensitive instructions and processing fee requests also elevated the overall phishing risk associated with the email.

5.2.8 Identified Phishing Indicators

No.	Phishing Indicator	Threat Level	Explanation
1.	Impersonal Greeting	Medium	Does not address the recipient using their actual name
2.	False Rewards & Offers	High	Promises immediate loan approval to attract victims
3.	Spoofed Sender Address	High	Uses an unofficial banking-related domain to imitate a financial institution
4.	Suspicious URLs	High	Contains a verification button that may redirect users to a fraudulent website
5.	Immediate Action Requests	High	Requests account verification and payment within 24 hours
6.	Confidentiality Requests	High	Advises users not to share the email details with others
7.	Fake Processing Fee Request	Critical	Asks the victim to pay a processing fee before receiving funds
8.	Credential Collection	Critical	Attempts to collect banking and personal information

5.2.9 Phishing Attack Mechanism

The attacker impersonates a financial institution by creating a professionally designed phishing email presenting a fake loan approval. The email attempts to create excitement and pressure by notifying the victim that a personal loan has already been approved.

The victim is required to verify banking information and pay a processing fee before the loan amount can be released. Once the victim provides sensitive information or transfers money, the attacker obtains financial and personal data that can be utilised for fraud and identity theft.

5.2.10 Potential Security Impact

- Financial fraud and monetary loss
- Exposure of banking and personal information
- Identity theft and credential compromise
- Loss of trust in financial communication systems

5.2.11 Security Recommendations

- Verify financial offers directly through official banking websites
- Avoid paying advance processing fees through email requests
- Confirm sender domains carefully before responding
- Avoid clicking suspicious verification links
- Report fraudulent financial emails to security teams immediately

5.3 Sample 3 - Netflix Billing Suspension Phishing Email

5.3.1 Email Type

Subscription Billing Phishing Scam

5.3.2 Suspicious Sender Address

support@netflix-billing.com

5.3.3 Email Subject

Payment Update Required

5.3.4 Threat Severity Assessment

Confirmed Phishing Email – High Risk

5.3.5 Phishing Email Sample

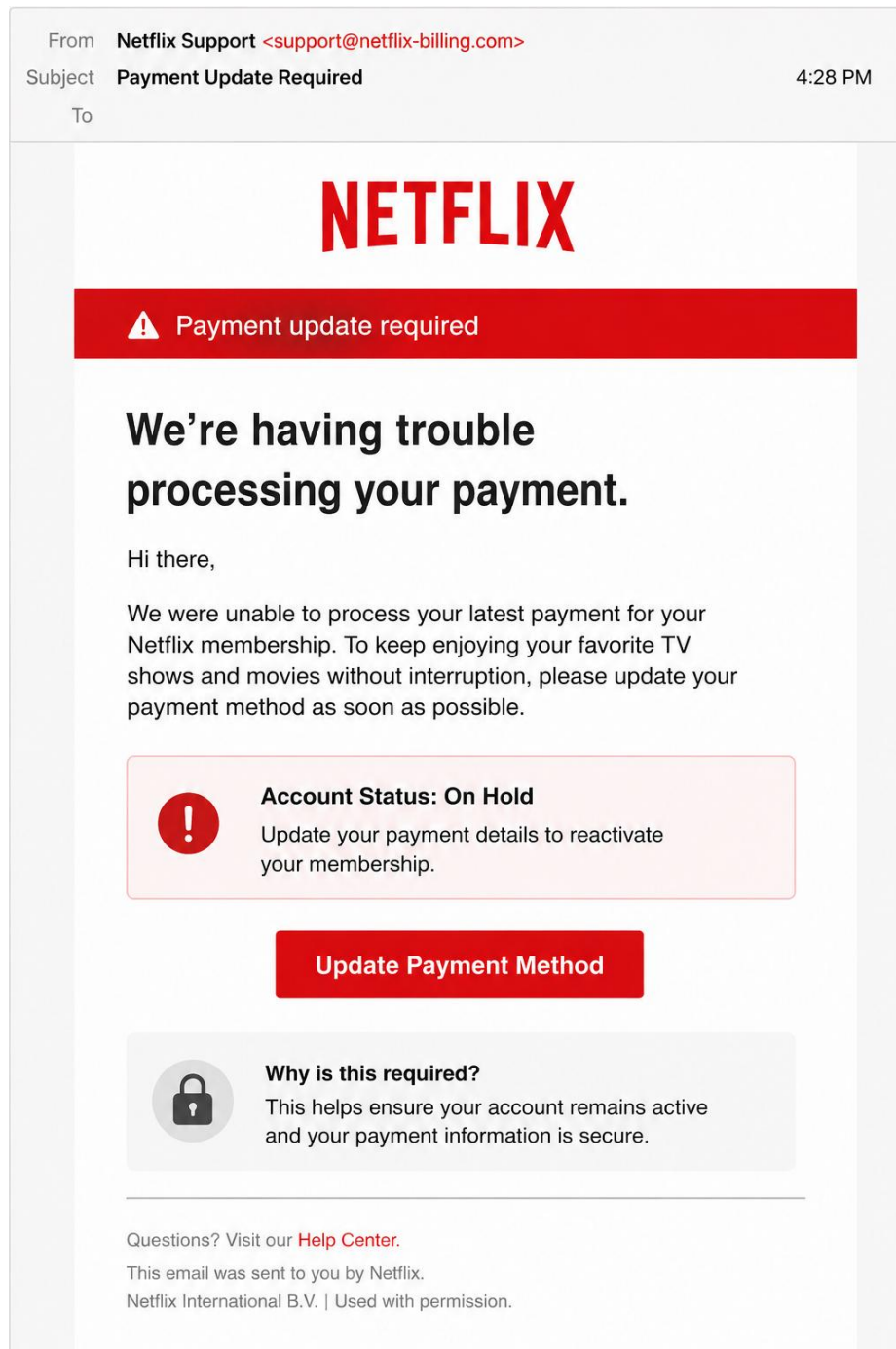


Figure 7: Netflix billing suspension phishing email sample

5.3.6 Email Header Analysis Using MXToolbox

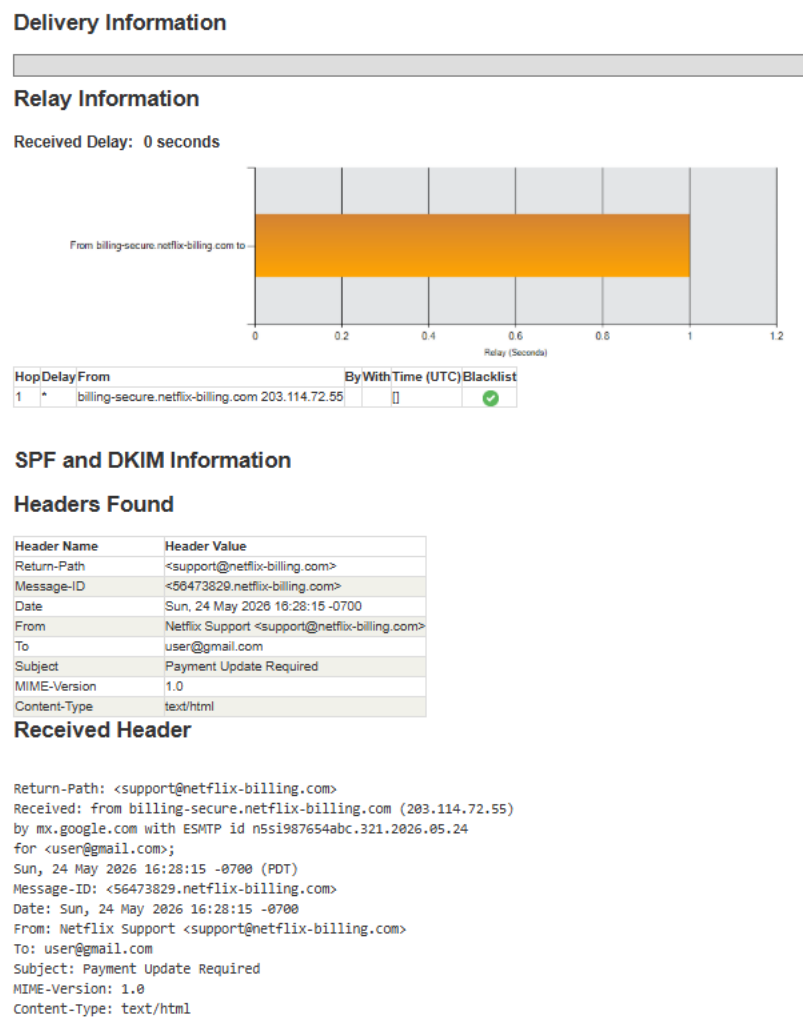


Figure 8: MXToolbox header analysis results for the Netflix phishing email

Explanation

The MXToolbox analysis detected suspicious sender infrastructure associated with the domain “netflix-billing.com”, which does not match the official Netflix domain structure. The routing information revealed that the email originated from an external server unrelated to official Netflix communication systems, indicating possible spoofing behaviour typically associated with phishing campaigns.

The analysis also uncovered suspicious sender information and external routing activity that further confirm the classification of the email as a phishing attempt created to steal payment and account credentials.

5.3.7 Email Header Analysis Using Google Admin Toolbox

Google Admin Toolbox	Messageheader	Help
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MessageId	56473829.netflix-billing.com
Created at:	5/25/2026, 3:28:15 AM GMT+4 (Delivered after)
From:	Netflix Support <support@netflix-billing.com>
To:	user@gmail.com
Subject:	Payment Update Required

#	Delay	From *	To *	Protocol	Time received
0		billing-secure.netflix-billing.com	→ [Google] mx.google.com	ESMTP	5/25/2026, 3:28:15 AM GMT+4

Figure 9: Google Admin Toolbox header analysis results for the phishing email

Explanation

The Google Admin Toolbox analysis uncovered suspicious sender domain information and unusual routing details inconsistent with official Netflix email infrastructure. The sender address attempted to copy the Netflix support system while leveraging a fake external domain.

The analysis further exhibited phishing characteristics commonly found in subscription billing scams designed to manipulate victims into submitting payment information and account credentials through fraudulent links.

5.3.8 Identified Phishing Indicators

No.	Phishing Indicator	Threat Level	Explanation
1.	Impersonal Greeting	Medium	Uses “Hi there” instead of the recipient’s actual name
2.	Immediate Action Requests	High	Pressures users to update payment information immediately
3.	Spoofed Sender Address	High	Uses unofficial domain “netflix-billing.com” to imitate Netflix
4.	Suspicious URLs	High	Contains a payment update button that may redirect users to a fake website
5.	Prevents Verification	High	Encourages users to use only the provided link instead of official channels
6.	Credential Collection	Critical	Attempts to collect billing details and account credentials
7.	Fake Processing Fee Request	Critical	Requests updated payment and subscription billing information
8.	Executive Impersonation	Critical	Uses Netflix branding and professional formatting to appear legitimate

5.3.9 Phishing Attack Mechanism

The attacker impersonates Netflix by crafting an expertly designed phishing email claiming there is an issue with the user's payment method. The email attempts to create pressure by informing the victim that their account is currently on hold and requires instant payment verification.

The victim is instructed to update billing information through a suspicious payment link. Once the victim submits payment credentials or login information, the attacker can gain unauthorised access to financial and personal account data for fraudulent activities.

5.3.10 Potential Security Impact

- Exposure of payment and billing information
- Unauthorised access to streaming accounts
- Financial fraud and identity theft
- Loss of sensitive personal information

5.3.11 Security Recommendations

- Verify payment issues directly through the official Netflix website
- Avoid clicking suspicious billing update links in emails
- Check sender domains carefully before responding
- Enable multi-factor authentication on streaming accounts
- Report suspicious subscription emails to security teams immediately

5.4 Sample 4 - Fake IT Security Update Phishing Email

5.4.1 Email Type

Malware Delivery / Fake Security Update Scam

5.4.2 Suspicious Sender Address

it.support@sort-IT-Out.net

5.4.3 Email Subject

Critical Security Update Required - Action Needed

5.4.4 Threat Severity Assessment

Confirmed Phishing Email – Critical Risk

5.4.5 Phishing Email Sample

Critical Security Update Required - Action Needed



IT <it.support@sort-IT-Out.net>
to me ▾

Hi there,

This is Mike, from IT Support at Sort IT Out. We've detected a critical security vulnerability affecting all company devices. You need to install this security patch immediately to prevent a potential breach.

Download and run the security update: <https://sort-IT-Out-updates.tech/patch.exe>

Please complete this within the next 2 hours. If you have any issues, just reply to this email.

Thanks,

Mike Thompson
IT Support, Sort IT Out

Figure 10: Fake IT security update phishing email sample

5.4.6 Email Header Analysis Using MXToolbox

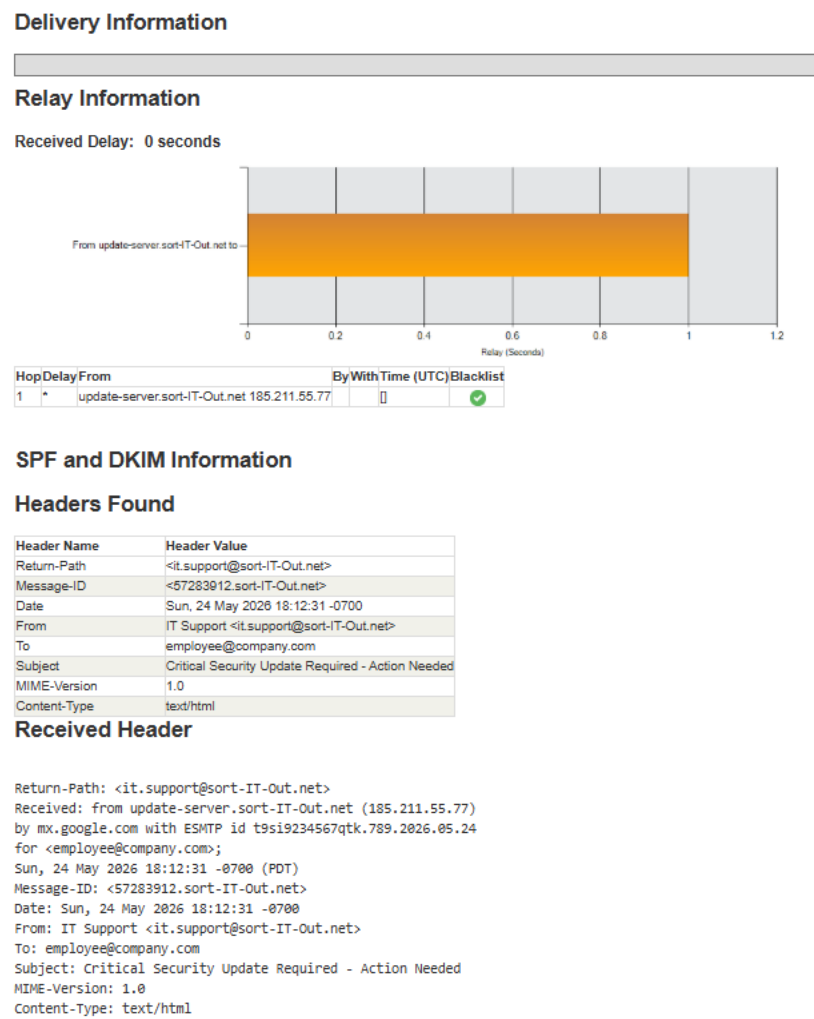


Figure 11: MXToolbox header analysis results for the fake IT security update email

Explanation

The MXToolbox analysis detected suspicious sender infrastructure associated with the domain “sort-IT-Out.net”, which does not match the official domain structure typically used by legitimate corporate IT departments. The routing information revealed that the email originated from an unknown external server, indicating possible spoofing activity commonly associated with malware delivery and phishing campaigns.

The analysis also uncovered suspicious sender details and external routing behaviour, which further support the classification of the email as a malicious phishing attempt designed to distribute harmful software.

5.4.7 Email Header Analysis Using Google Admin Toolbox

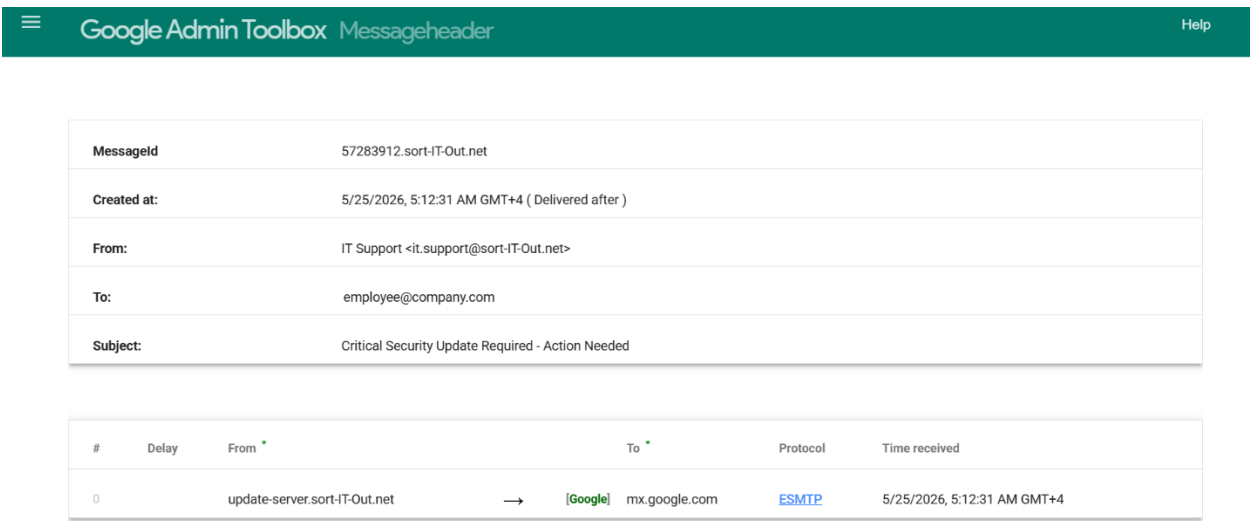


Figure 12: Google Admin Toolbox header analysis results for the phishing email

Explanation

The Google Admin Toolbox analysis identified suspicious sender domain information and abnormal routing details inconsistent with legitimate organisational IT communication systems. The sender address attempted to mimic an internal IT support department while employing a misleading external domain.

The analysis further demonstrated phishing characteristics commonly observed in malware delivery campaigns designed to trick employees into downloading malicious files disguised as urgent security updates.

5.4.8 Identified Phishing Indicators

No.	Phishing Indicator	Threat Level	Explanation
1.	Impersonal Greeting	Medium	Uses “Hi there” instead of addressing the employee by name
2.	Immediate Action Requests	High	Pressures users to install the security update immediately
3.	Spoofed Sender Address	High	Uses unofficial domain “sort-IT-Out.net” to imitate a trusted IT department
4.	Suspicious URLs	High	Contains a suspicious external download link ending with “patch.exe”
5.	Prevents Verification	High	Encourages users to use only the provided link instead of verifying with the IT department
6.	Confidentiality Requests	High	Advises users to reply only through the email to avoid external verification
7.	Credential Collection	Critical	Malware installation may lead to credential theft and unauthorised access

8.	Executive Impersonation	Critical	Pretends to be an official organisational IT support notification to gain trust
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5.4.9 Phishing Attack Mechanism

The attacker poses as an organisational IT support department by creating a fake security alert email claiming that company devices are at risk to a critical security threat. The email attempts to create urgency and fear by warning users about a possible system compromise if immediate action is not taken.

The victim is instructed to download and execute a security patch from an external link. Once the malicious file is downloaded and executed, malware may be installed on the victim's device, potentially allowing attackers to steal data, monitor activity, or gain unauthorised access to organisational systems.

5.4.10 Potential Security Impact

- Malware infection across company devices
- Unauthorised access to organisational systems
- Data theft and credential compromise
- Network disruption and operational downtime

5.4.11 Security Recommendations

- Never download software updates from email links
- Verify IT security alerts directly with the official IT department

- Avoid executing unknown executable files from external sources
- Use endpoint protection and antivirus software
- Report suspicious security update emails immediately

6. Prevention Guidelines for Users

6.1 Prevention Tips for Users

- Always verify the sender's email address before opening links or attachments.
- Avoid clicking suspicious links received through emails or messages.
- Check whether the domain name matches the official company website.
- Never share passwords, OTPs, or banking information through email.
- Be cautious of urgent messages demanding immediate action.
- Confirm security alerts directly through official websites or support teams.
- Avoid downloading executable files or software updates from email links.
- Enable multi-factor authentication (MFA) on important accounts.
- Keep antivirus software and operating systems updated regularly.
- Report suspicious emails to the organisation's IT or security department immediately.

6.2 Do's and Don'ts for Employees

Do's

- Verify suspicious emails with the official sender before responding.
- Check links carefully by hovering over URLs before clicking them.
- Use strong and unique passwords for different accounts.
- Report phishing emails immediately to the IT/security team.
- Keep systems and security software updated regularly.
- Follow organisational cybersecurity policies and awareness guidelines.
- Use multi-factor authentication whenever possible.

Don'ts

- Do not click unknown links or download suspicious attachments.
- Do not share passwords, OTPs, or confidential information through email.
- Do not trust emails that create urgency or fear without verification.
- Do not install software updates from unofficial sources.
- Do not respond to suspicious financial or account verification requests.
- Do not ignore phishing warning signs such as fake domains or generic greetings.
- Do not forward suspicious emails to other employees without reporting them first.

7. Conclusion

This report provided a detailed analysis of multiple phishing email samples designed to imitate trusted organisations and manipulate users into revealing sensitive information, downloading malicious files, or making fraudulent payments. The investigation found several common phishing characteristics, including spoofed sender domains, urgent language, suspicious links, fake security notifications, and credential collection attempts.

Email header analysis using MXToolbox and Google Admin Toolbox further showed how phishing emails often come from suspicious external servers and misleading domains rather than legitimate organisational infrastructure. The findings demonstrated that phishing attacks continue to depend heavily on social engineering tactics to exploit user trust and create panic or urgency.

The report also highlighted the value of cybersecurity awareness, employee training, and proper email verification practices in reducing phishing-related risks. By following recommended security measures such as verifying sender identities, avoiding suspicious links, enabling multi-factor authentication, and reporting malicious emails promptly, both individuals and organisations can significantly decrease the risk of successful phishing attacks.

Overall, the analysed samples were classified as high-risk or critical phishing threats due to their potential to cause credential theft, financial fraud, malware infections, and unauthorised system access. Regular security awareness and proactive monitoring remain crucial for protecting organisational and personal information from emerging phishing campaigns.